

Evaluating Robustness of Lightweight Object Detectors under Compound Variability in UAV Aerial Imagery

PhD Research Progress · First Study Summary

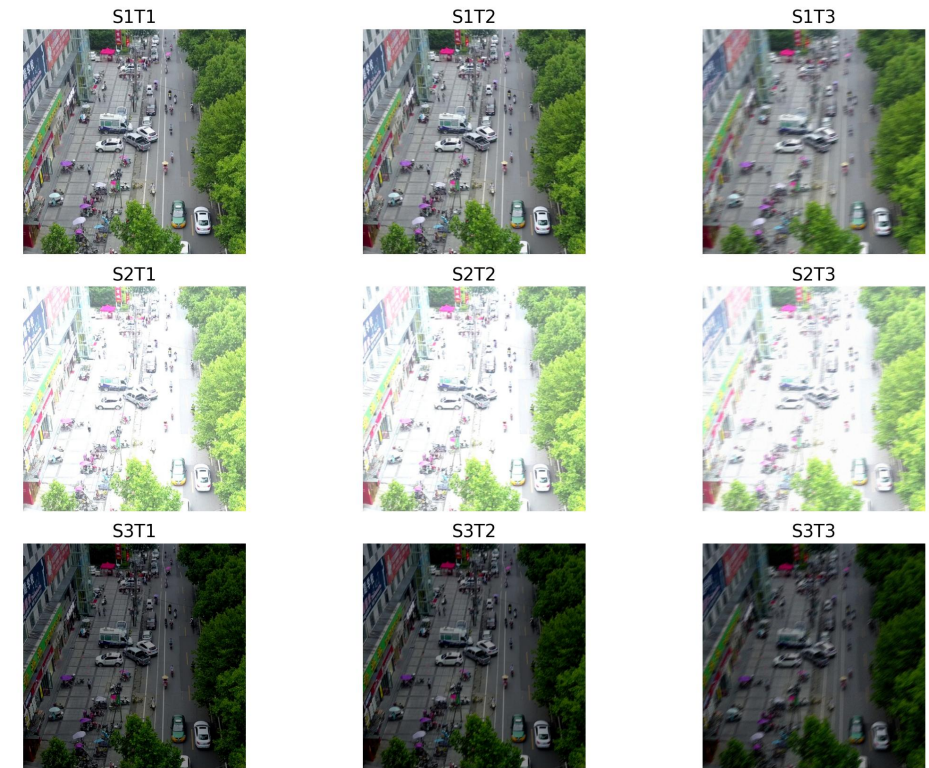
3×3 degradation

640 vs 1280

ZS vs FT

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Main idea: separate clean accuracy, target-size effects, domain adaptation, and corruption resistance.

Why this problem matters

UAV detection is limited by small target size and image degradation, not only by clean-image model accuracy.

Small targets

VisDrone contains dense aerial scenes where many objects occupy only a small image area. Downsampling can remove edges and localization cues needed for tiny/small targets.

Acquisition degradation

Illumination changes, camera motion and image noise can erase the same weak visual cues that small-object detectors depend on.

Clean $\neq$ operational robustness

A model that ranks well on clean images can still fail sharply under corrupted inputs. Robustness therefore needs separate measurement.

Research gap

Existing aerial studies often emphasize clean small-object accuracy or propose new architectures. This study asks a complementary question: how do degradation, resolution, fine-tuning, and model generation interact in a controlled evaluation?

The goal is diagnosis: understand when accuracy gains translate — or do not translate — into robustness.

Selected background: Zhu et al., IEEE TPAMI 2022 (VisDrone); Hendrycks & Dietterich, ICLR 2019; Michaelis et al., arXiv 2019.

Final research framework: four controlled questions

The study is organized around four factors; supporting models do not create additional research questions.

1 · Compound degradation

A fixed 3×3 matrix crosses illumination changes with motion blur / noise to expose difficult condition combinations.

2 · Resolution × target size

The same YOLOv8s-FT and YOLO26s-FT checkpoints are evaluated at 640 and 1280 to isolate inference-resolution effects.

3 · Domain adaptation

COCO-pretrained zero-shot and VisDrone fine-tuned checkpoints are compared on the same six shared classes.

4 · Detector generation

YOLOv8s vs YOLO26s is the primary generation comparison; YOLO11 and nano models are supporting evidence.

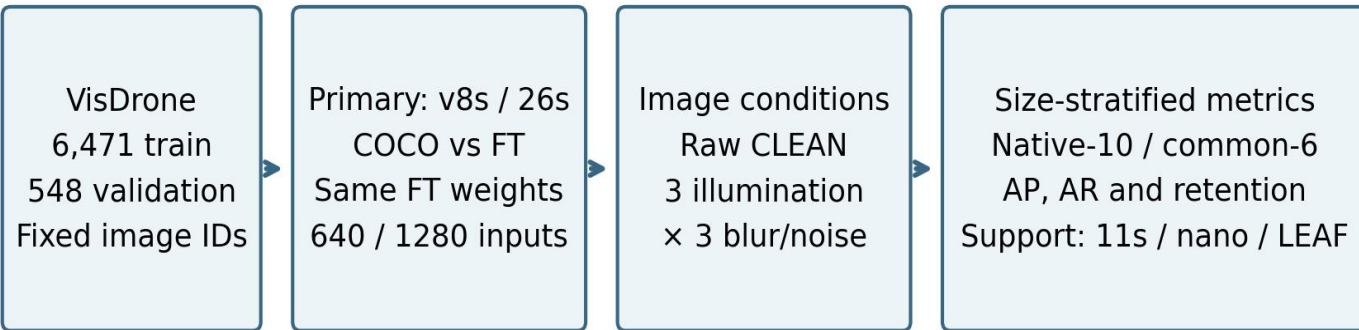
Controlled design principle

Keep checkpoint, label space, model capacity and evaluation protocol explicit so that one factor is not accidentally interpreted as another.

Primary comparison: YOLOv8s ↔ YOLO26s · Supporting: YOLO11, nano models, external LEAF-YOLO-N.

Dataset, models and evaluation workflow

One benchmark, fixed checkpoints, explicit label spaces, genuine COCO-style metrics.



Dataset

VisDrone2019-DET · 6,471 train + 548 validation images

Primary models

YOLOv8s and YOLO26s (1280 training). YOLO11s and nano models provide supporting comparisons.

Training

100 epochs · one seed · shared high-level schedule · best validation checkpoint

Evaluation

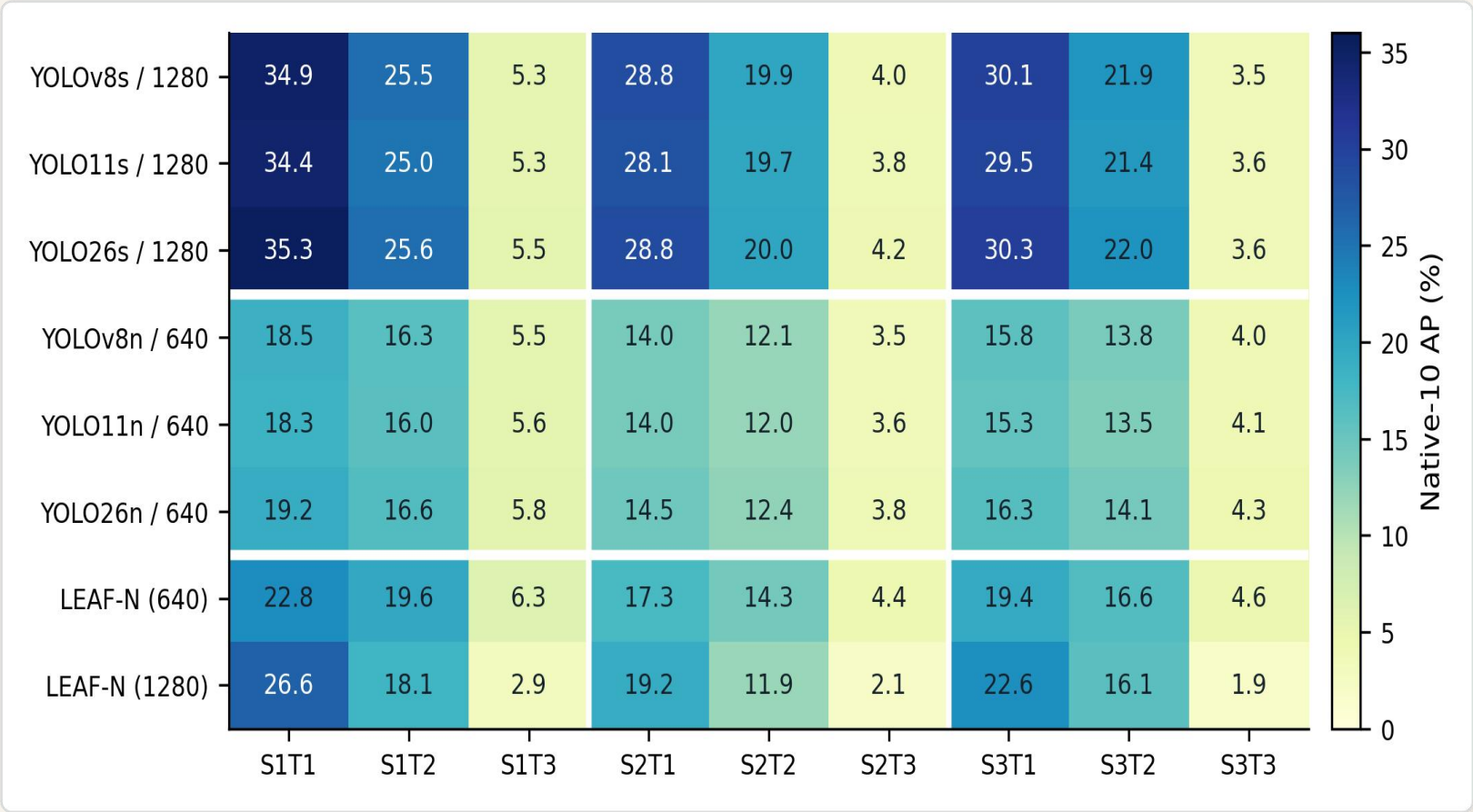
COCO AP50:95 + AP50 + AR · per-class and fixed size bins · native-10 vs common-six kept separate

Robustness

Mean corrupted AP · retention = corrupt mean / raw CLEAN · worst-condition AP recorded

Result 1 · Compound degradation reveals the main failure regime

The 3×3 matrix makes the difficult combinations visible instead of hiding them in one average score.



Primary failure

For both YOLOv8s and YOLO26s at 1280, the lowest AP occurs at S3T3.

S3T3

Underexposure + severe motion blur + Gaussian noise reduces AP to roughly one tenth of clean performance.

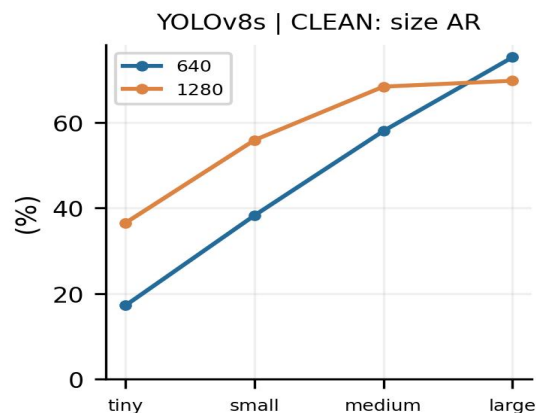
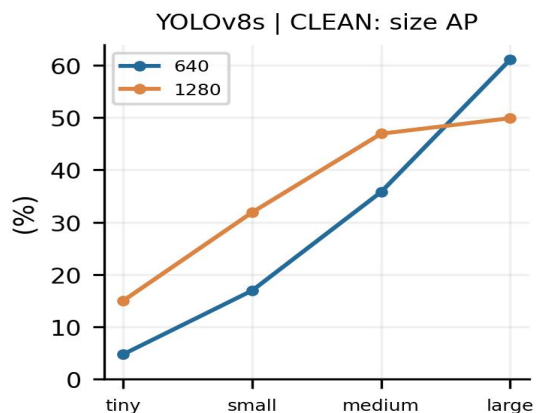
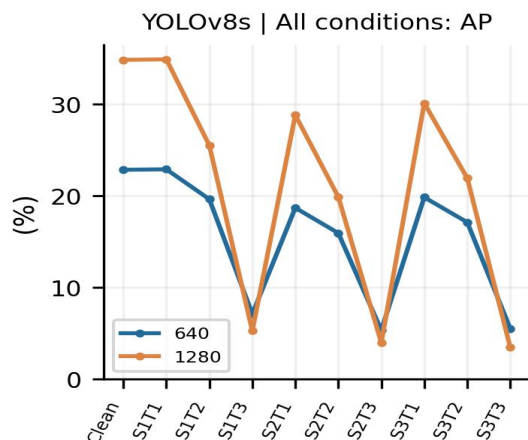
Interpretation

The corruption effect is much larger than the small clean-score difference between recent detector generations.

Main message: selecting a detector from a clean leaderboard does not tell us how it behaves under difficult UAV image conditions.

Result 2 · Higher inference resolution helps small targets — but not uniformly

Same checkpoint at 640 and 1280: no retraining and no model-capacity change.

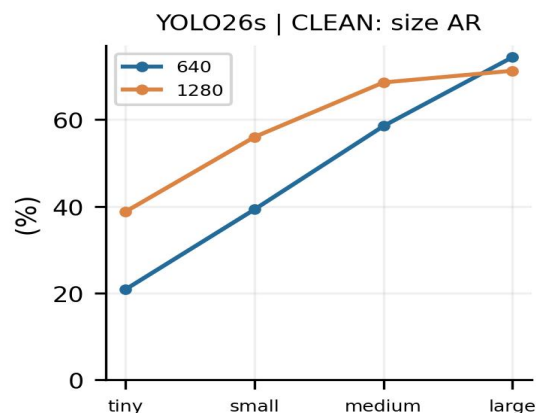
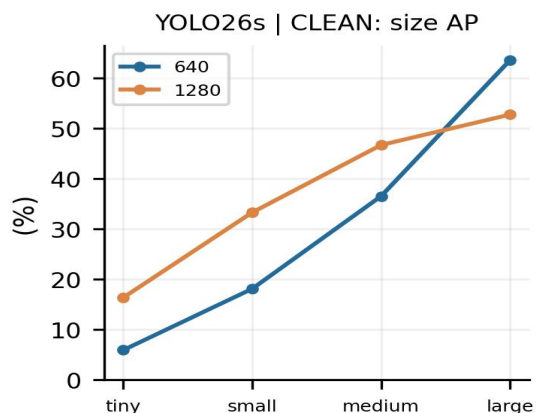
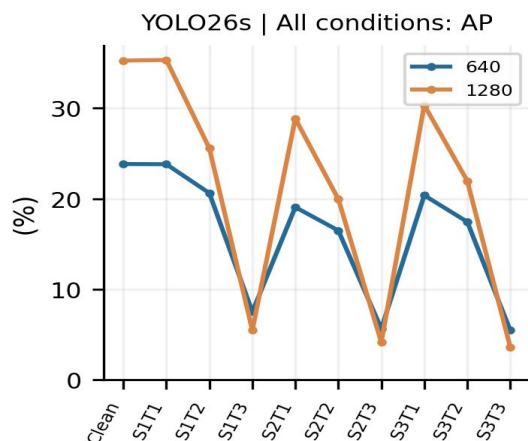


Small AP gain

YOLOv8s: +14.93 pp
YOLO26s: +15.17 pp

Tiny AR gain

YOLOv8s: +19.27 pp
YOLO26s: +17.92 pp



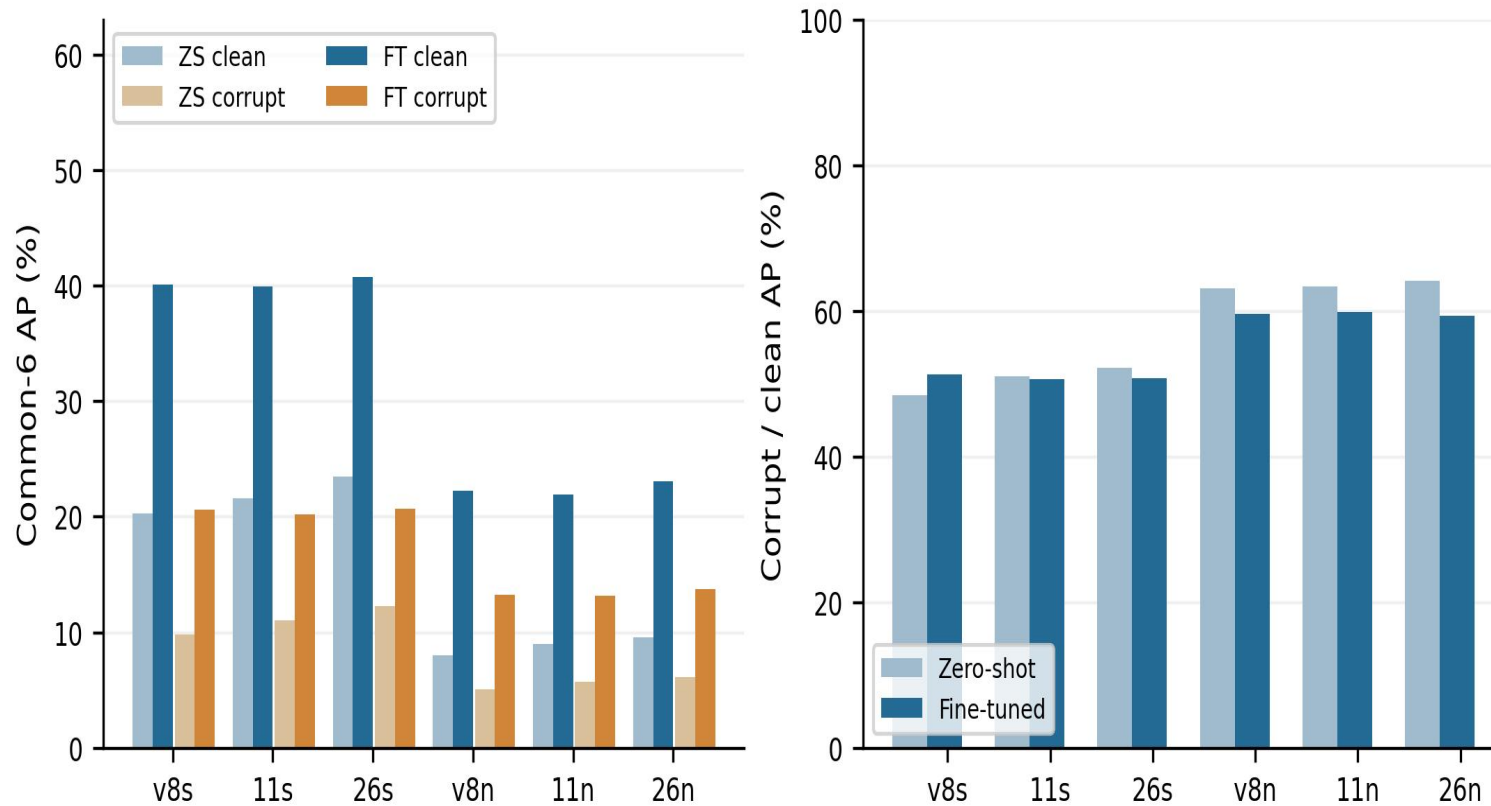
Trade-off

Large-object AP decreases at 1280 for both primary models; retention also falls even though absolute corrupted AP increases.

Resolution is an operating choice: it improves small-target evidence, but the robustness and size trade-offs must be measured.

Result 3 · Fine-tuning improves absolute AP, not necessarily proportional robustness

The common-six task keeps the target vocabulary identical before and after VisDrone fine-tuning.



YOLOv8s

Clean: 20.28 → 40.07%
Corrupt mean: 9.83 → 20.57%
Retention: 48.5 → 51.3%

YOLO26s

Clean: 23.49 → 40.76%
Corrupt mean: 12.27 → 20.71%
Retention: 52.2 → 50.8%

Why it matters

A detector can become much more accurate under corruption while retaining a smaller fraction of its clean score.

Use both absolute corrupted AP and retention; “fine-tuning improves robustness” is incomplete unless robustness is defined.

Result 4 · YOLO26s improves clean small-target accuracy more than corrupted robustness

Primary generational comparison at 1280.

Metric comparison

Clean AP



Tiny-object AP



Mean corrupted AP



Difference (YOLO26s – YOLOv8s) **+0.44 pp clean · +1.36 pp tiny · ~+0.13 pp corrupted mean**

What improved most?

The clean tiny-object gain is larger than the overall clean gain.

What did not change much?

Mean corrupted AP remains almost tied between YOLOv8s and YOLO26s.

Interpretation

Newer ≠ automatically more robust. The result is family-level evidence, not a component ablation.

The strongest YOLO26s signal is clean tiny-target accuracy — not a large robustness advantage.

How this study is positioned relative to recent literature

The study complements method papers by isolating operating factors and failure modes.

VisDrone benchmark

Zhu et al.
IEEE TPAMI, 2022

Defines the aerial detection benchmark and highlights viewpoint, scale variation and motion blur challenges.

Small-object methods

MFFSODNet (2024)
LEAF-YOLO (2025)

Improve small-object representation and lightweight deployment on aerial imagery.

Robust architecture

YOLO-ROSS
Remote Sensing, 2026

Proposes a new YOLOv11-based detector for controlled corruption robustness and cross-domain transfer.

This study

Diagnostic factor study

3×3 illumination–motion matrix
Same-checkpoint 640/1280
ZS→FT analysis
YOLOv8s→YOLO26s

Complementary contribution: understand robustness trade-offs before designing the next degradation-aware detector.

Refs: Jiang et al., IEEE TIM 2024; Nghiem et al., ISWA 2025, doi:10.1016/j.iswa.2025.200484; Zhong et al., Remote Sens. 2026, 18(17), 2997, doi:10.3390/rs18172997.

Main takeaways and next research step

The current study is complete enough to freeze the experiment scope.

1 · Failure is condition-dependent

Severe underexposure + blur/noise causes a much larger performance loss than the small differences between recent detector generations.

2 · Resolution has a real small-target effect

1280 preserves more small-target information, but large-object AP and proportional retention can move in the opposite direction.

3 · Fine-tuning changes accuracy and retention differently

Fine-tuning is valuable in absolute AP, but robustness ratios need separate interpretation.

4 · Newer is not automatically more robust

YOLO26s shows a useful clean tiny-object gain, while its mean-corrupted advantage over YOLOv8s is small.

Next research direction

Move from “understand the failure” to “improve the failure”: a lightweight degradation-aware small-object detector, followed by cross-dataset / real UAV inspection validation.

For the current study: keep the scope frozen, complete a paper as soon as possible.